Getting started in Research and Analytics

Session 5: Regression

Regression

In this session, we'll cover Linear and Logistic Regression.

We can answer: "What is regression" & "what are my chances of surviving the Titanic"

Why should we learn this?

To get a variable's effect that controls for other variables.

To examine cause-and-effect and promote better practice.

Linear Regression

Background and steps

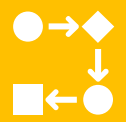
Motivation for multiple linear regression



There are 2 approaches in testing hypotheses : 1) Balance our sample on all important things and then test (e.g., correlation or t-test) or 2) gather data on all important things, then balance mathematically when we test.



#1 is called an “experiment” and #2 is an “observational study”. Observational studies work well when randomization is not possible.



Regression allows us to simultaneously test relationships, reducing Type-I errors. Otherwise, we would need to do separate tests.

Linear Regression (Ordinary Least Squares)

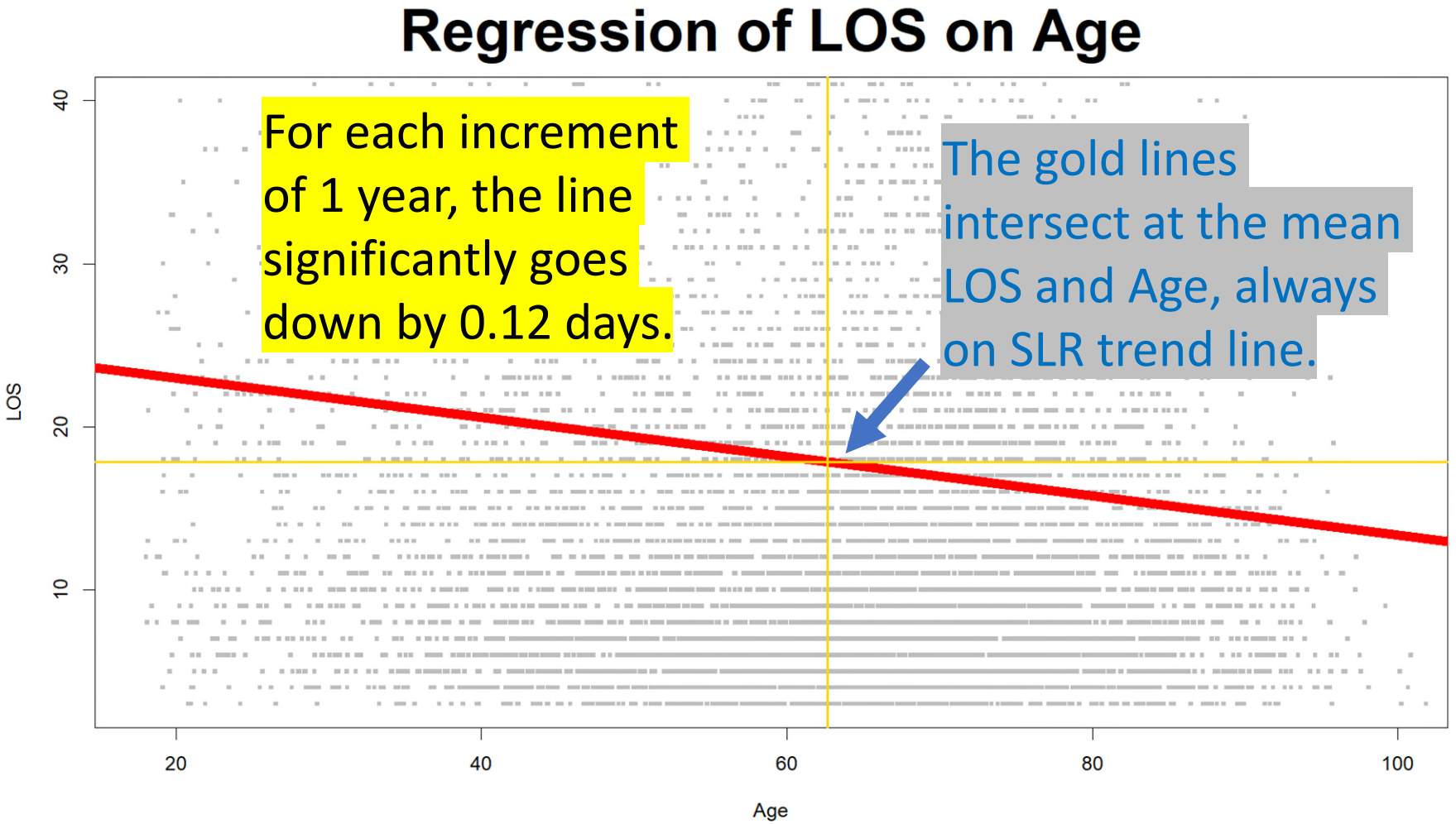
- Regression: Regressing the outcome on one (simple) or more predictors (multiple).
- Linear: Refers to a line that best represents the association between the outcome and the predictor (Y ~ X).
- Outcomes: In ordinary least squares, the outcome is continuous. These are variables with many values (e.g., length of stay, cost, etc.). Predictors can be continuous or categorical variables.
- We can use regression for 3 purposes: 1) Hypothesis testing, 2) effect size estimation—e.g., cost savings from intervention, and 3) predictive modeling—e.g., predicted LOS based on patient info.

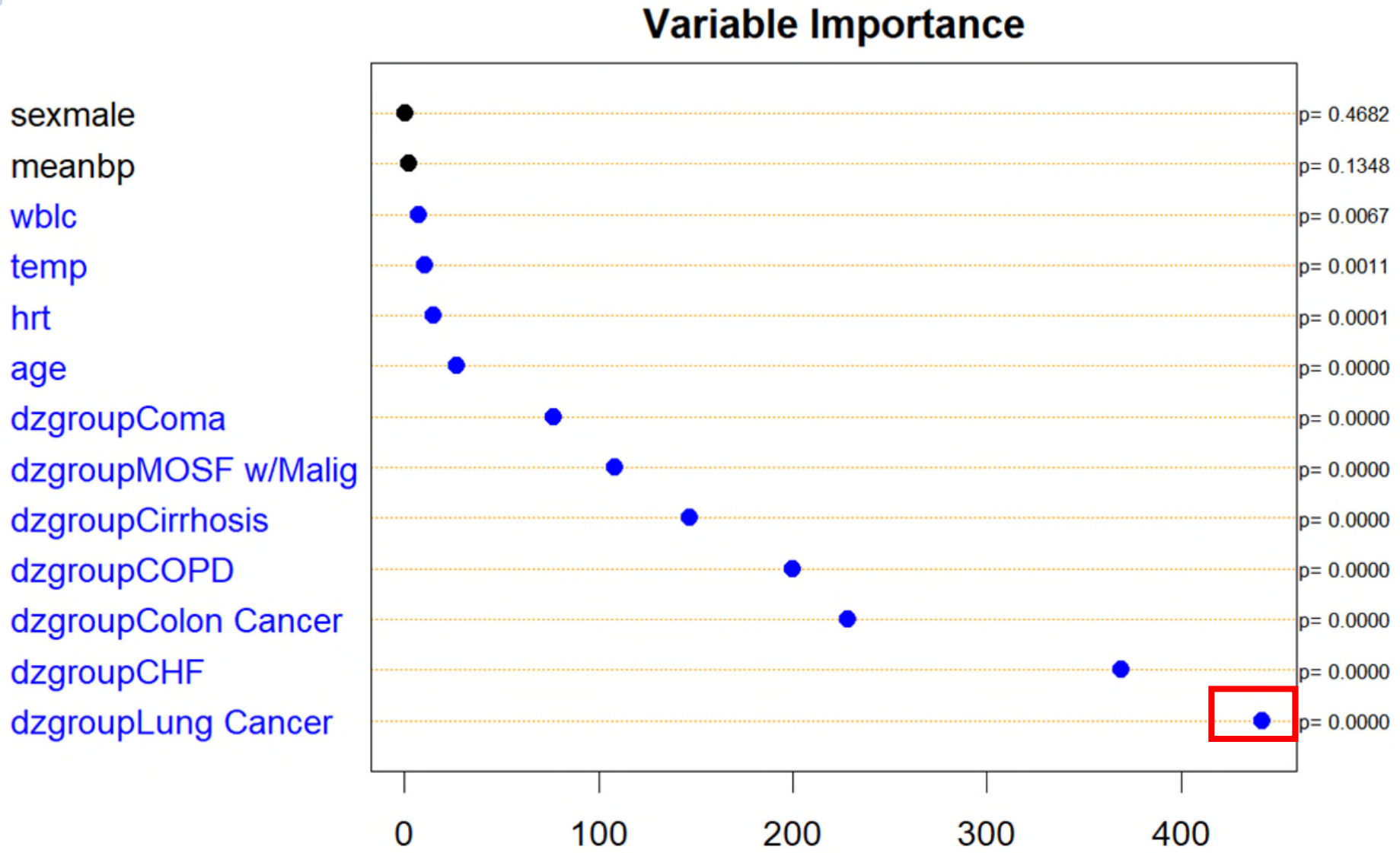


Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments (SUPPORT)

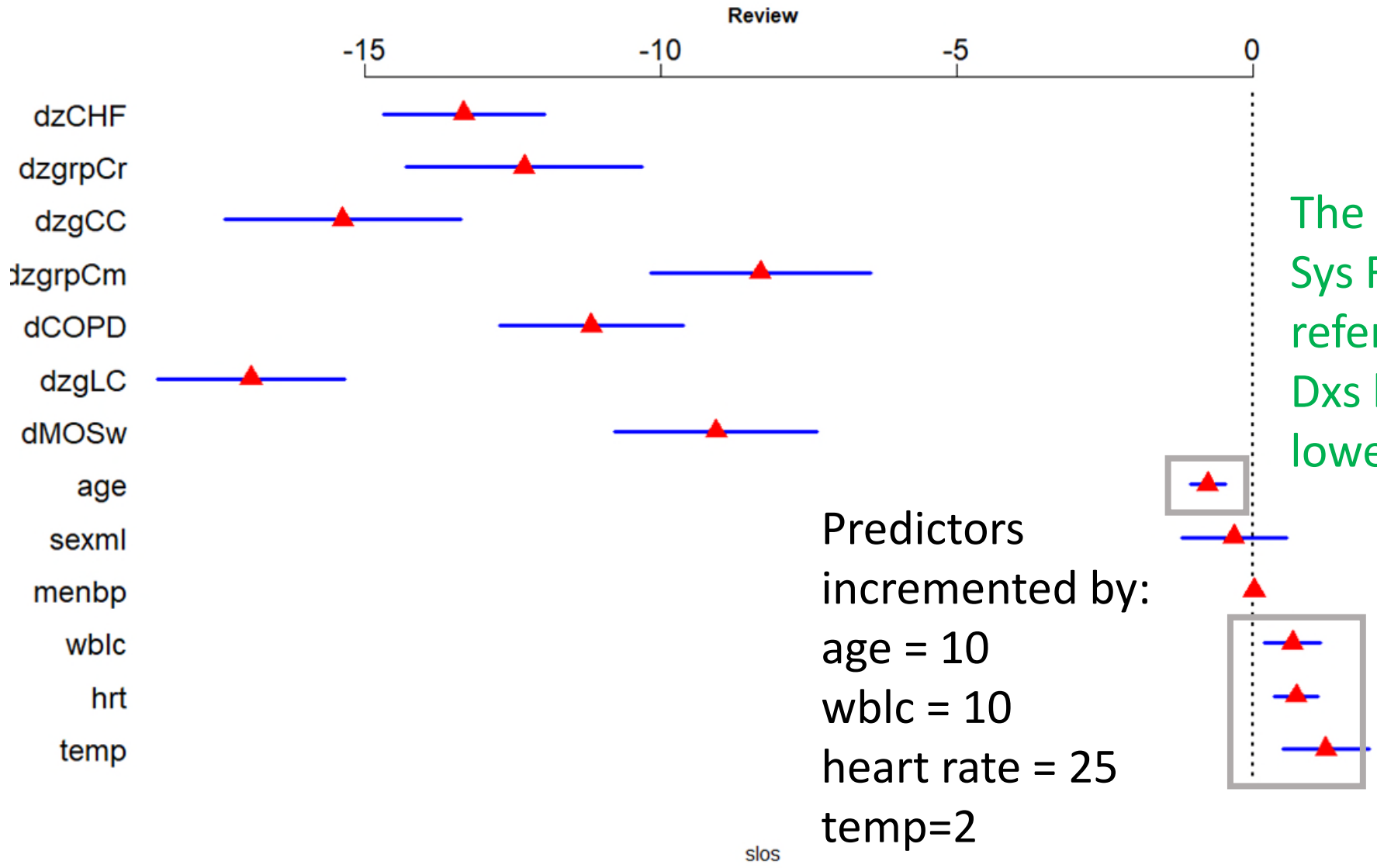
- Study to examine the outcomes of seriously ill hospitalized patients.
- Data collected from 1991-1994. N= 9,105
- The outcome is length of stay (LOS).

Simple Linear Regression





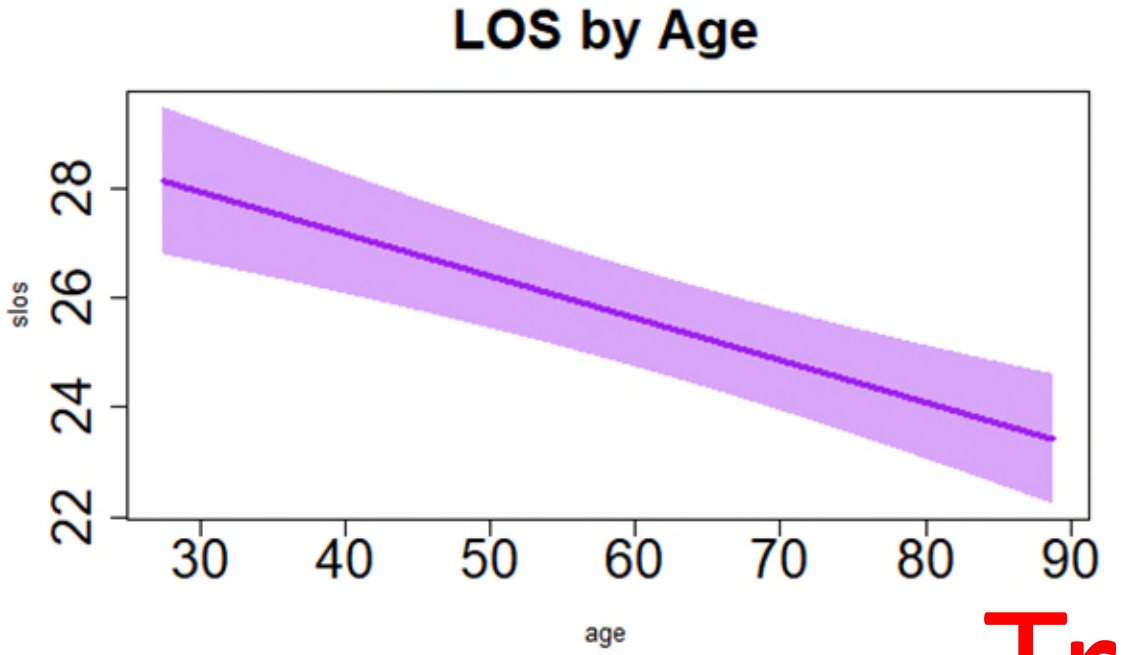
Most significant

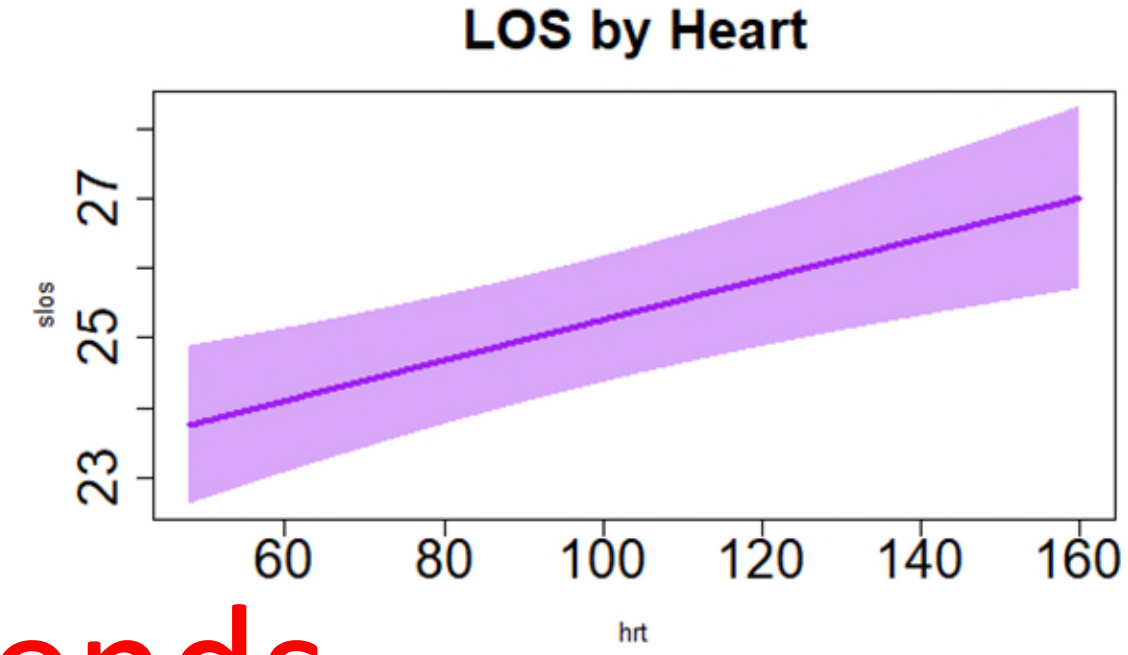


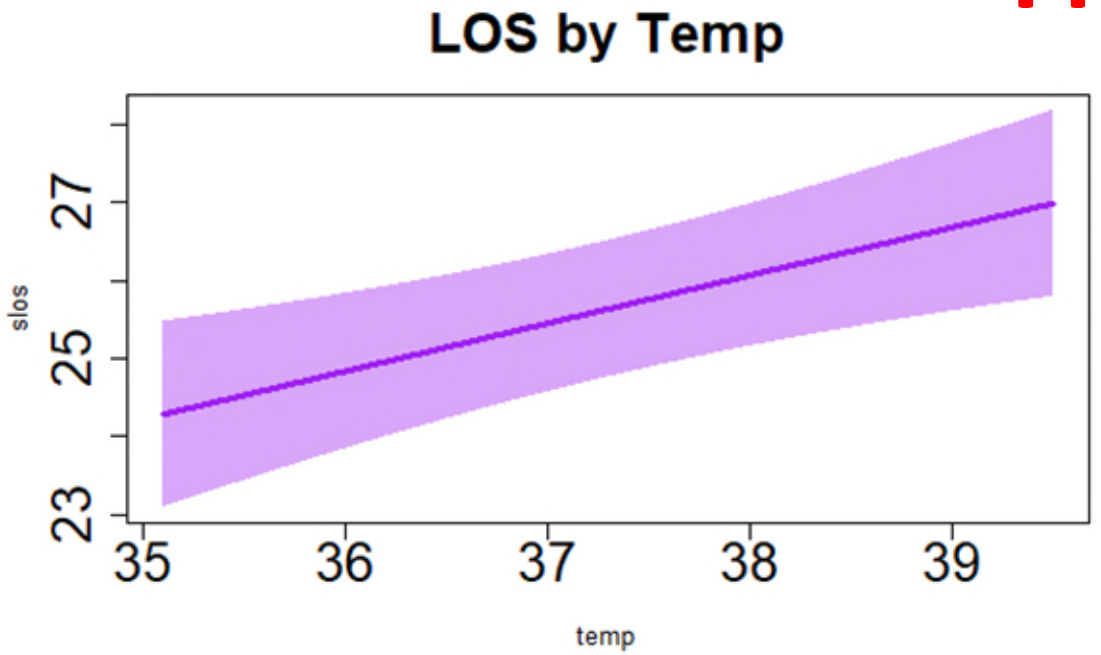
The ARF/Multi Organ Sys Failure is the Dx reference, all other Dxs have significantly lower LOS

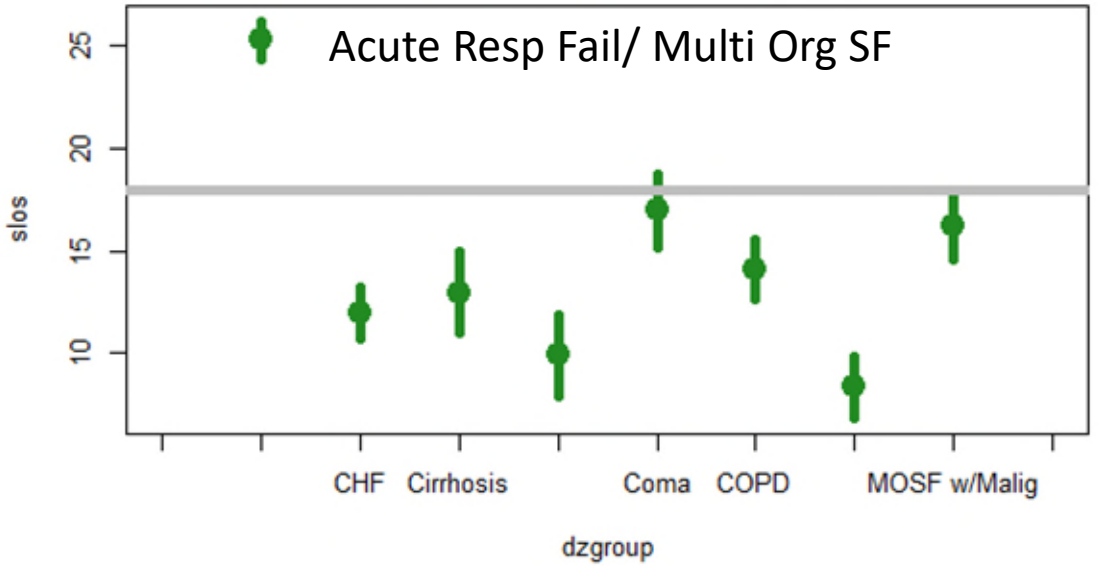
Predictors incremented by:
age = 10
wblc = 10
heart rate = 25
temp=2

Trends


LOS by Age
slos
age
22
24
26
28
30
40
50
60
70
80
90


LOS by Heart
slos
hrt
23
25
27
60
80
100
120
140
160


LOS by Temp
slos
temp
23
25
27
35
36
37
38
39


LOS by Disease
Acute Resp Fail/ Multi Org SF
slos
dzgroup
10
15
20
25
CHF
Cirrhosis
Coma
COPD
MOSF w/Malig

Logistic Regression

Background and steps

What is Logistic Regression?

- Logistic Regression is a “general linear model” for estimating the relationships (coefficients) between the independent and dependent variables (or outcomes) with only 2 levels (yes or no).
- Some variables may increase our chances while others decrease it.
- Logistic Regression is similar to Linear Regression but for binary outcomes.
- It’s a probability model (e.g., no normality assumption). Because probabilities can be converted into odds, we also get Odds Ratios in our results. E.g., women are 12 times more likely than men to survive the Titanic.

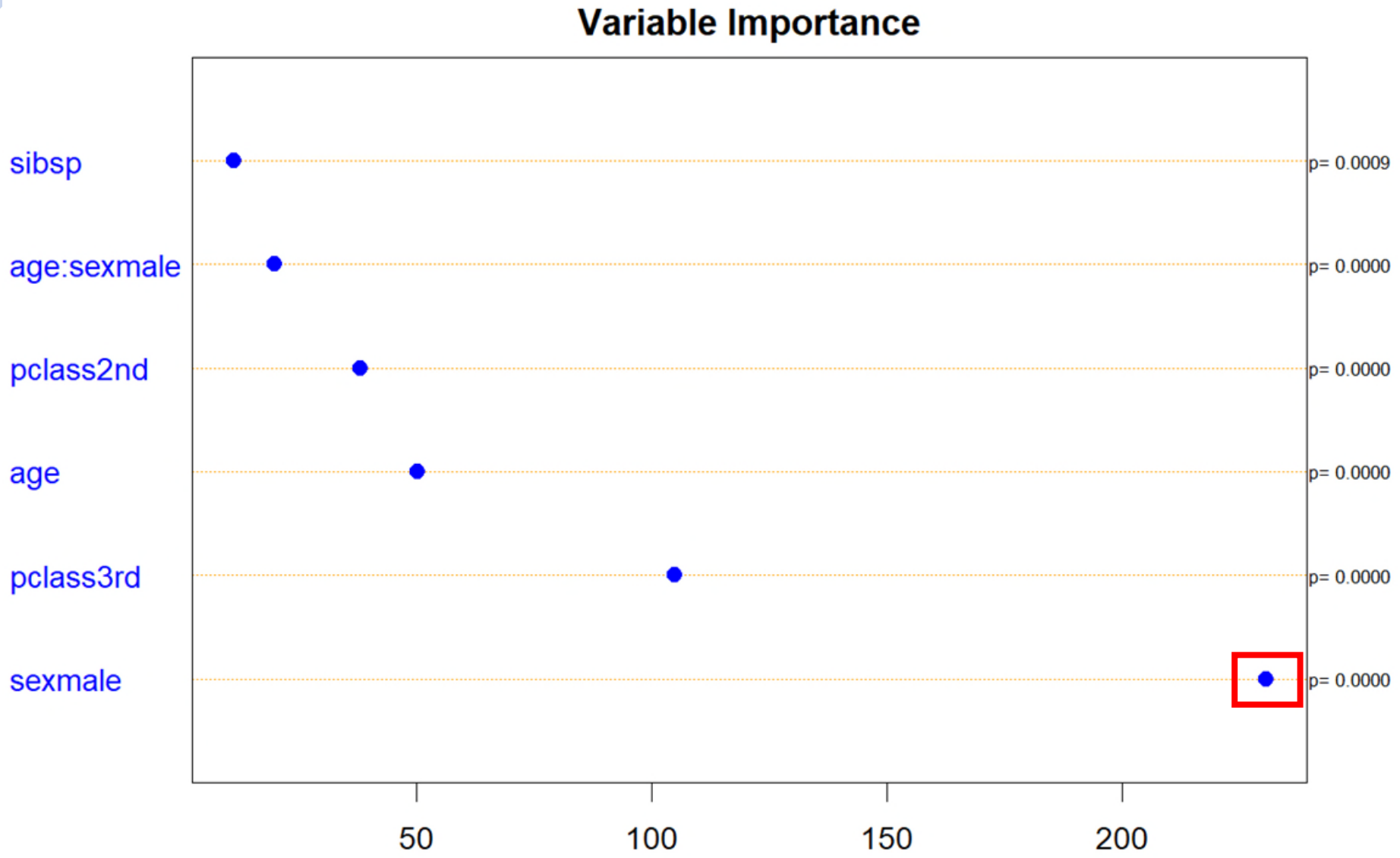
Conceptual model: Survival = Age * Gender + Passenger Class + siblings/spouses

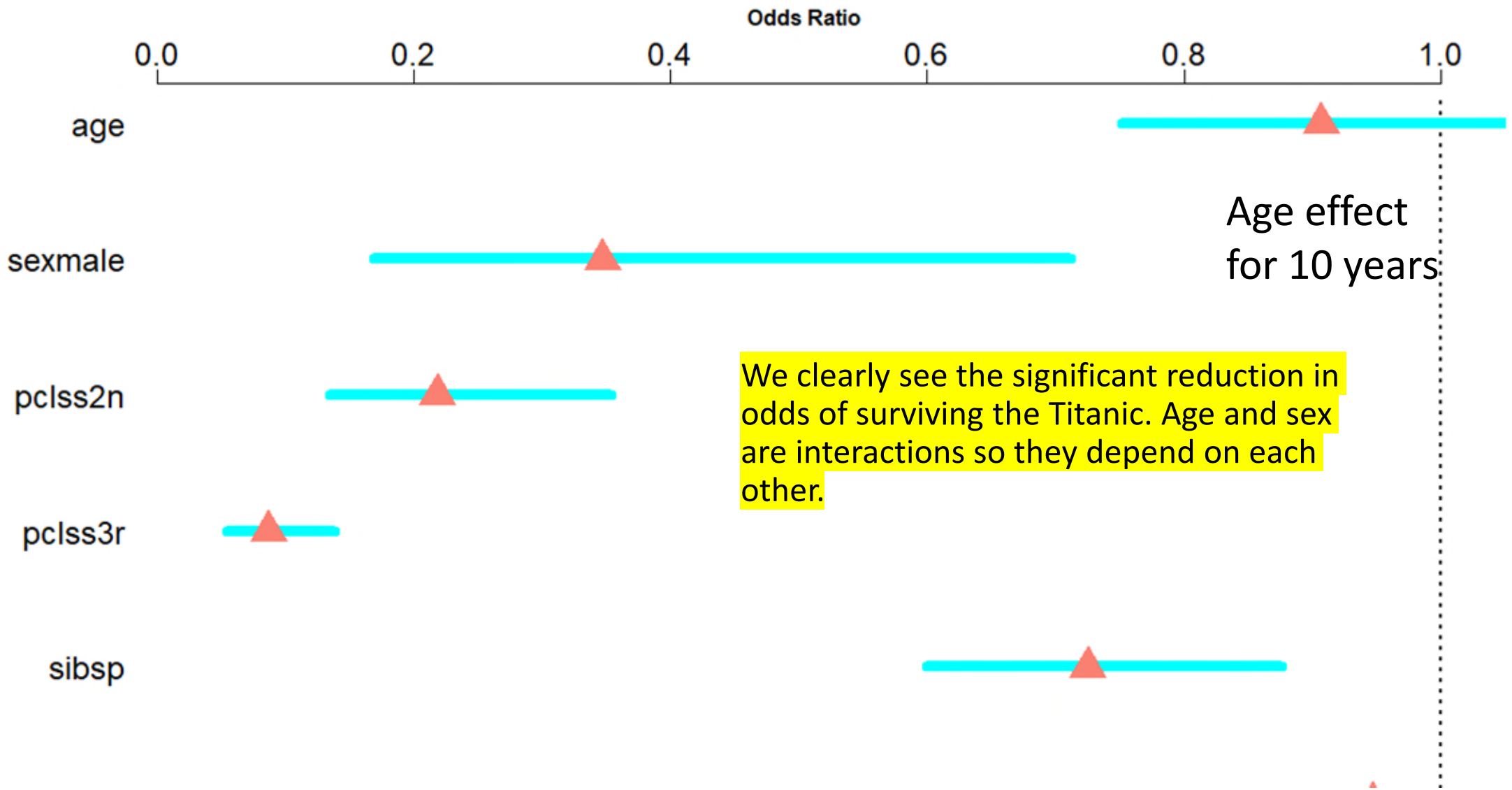
Surviving the Titanic (N = 1,046)

Millvina Dean (the youngest and last survivor of the Titanic)

- It really was a case of women-and-children first but we hear her say that passenger class made a very big difference. *We can test these hypotheses*.
- (Part 2) Millvina Dean: Inspiring Interview with Titanic's Youngest Survivor 🚢💪 #Titanic #History

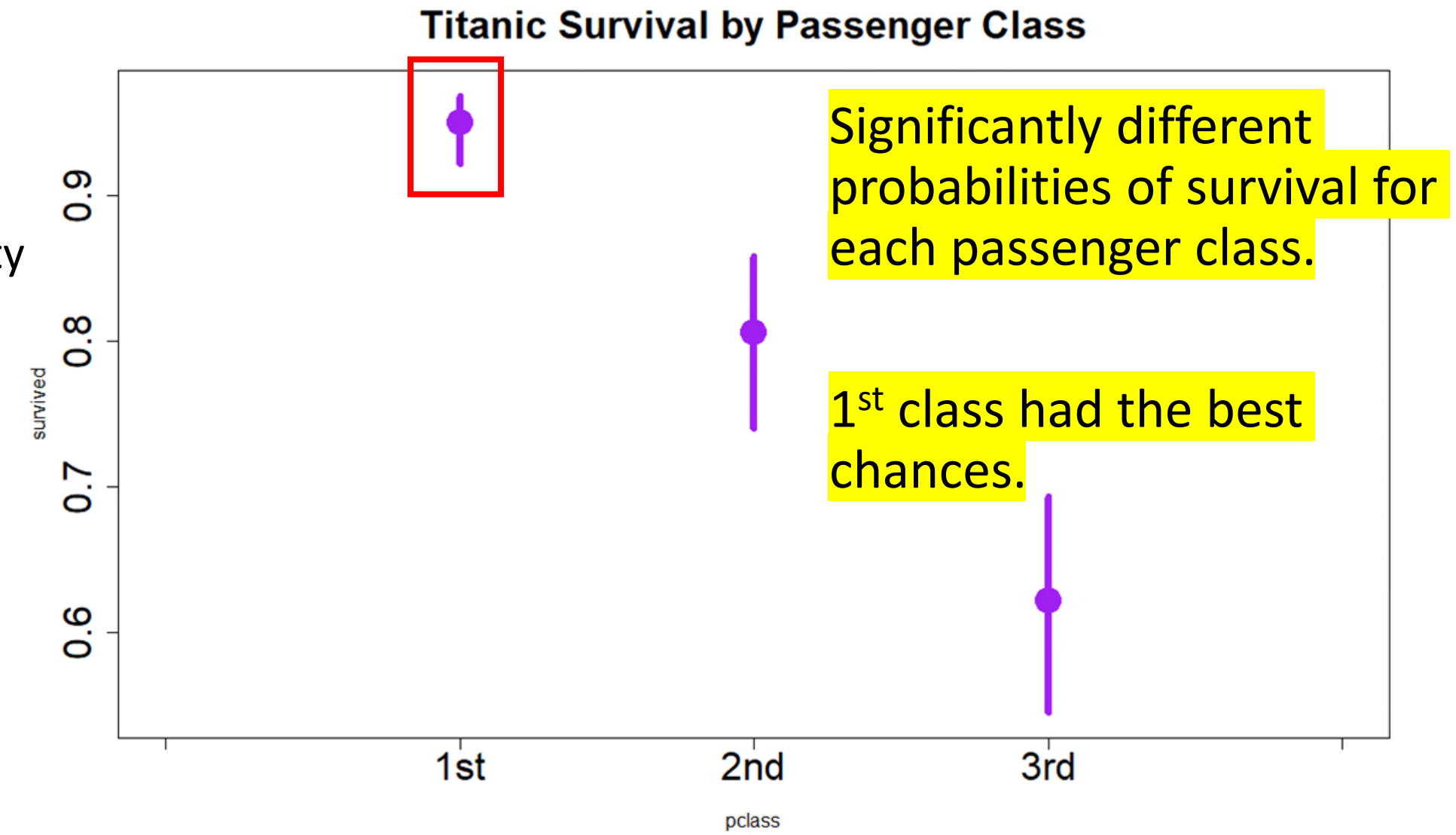



Variable Importance
sibsp
age:sexmale
pclass2nd
age
pclass3rd
sexmale
p= 0.0009
p= 0.0000
p= 0.0000
p= 0.0000
p= 0.0000
p= 0.0000
50
100
150
200



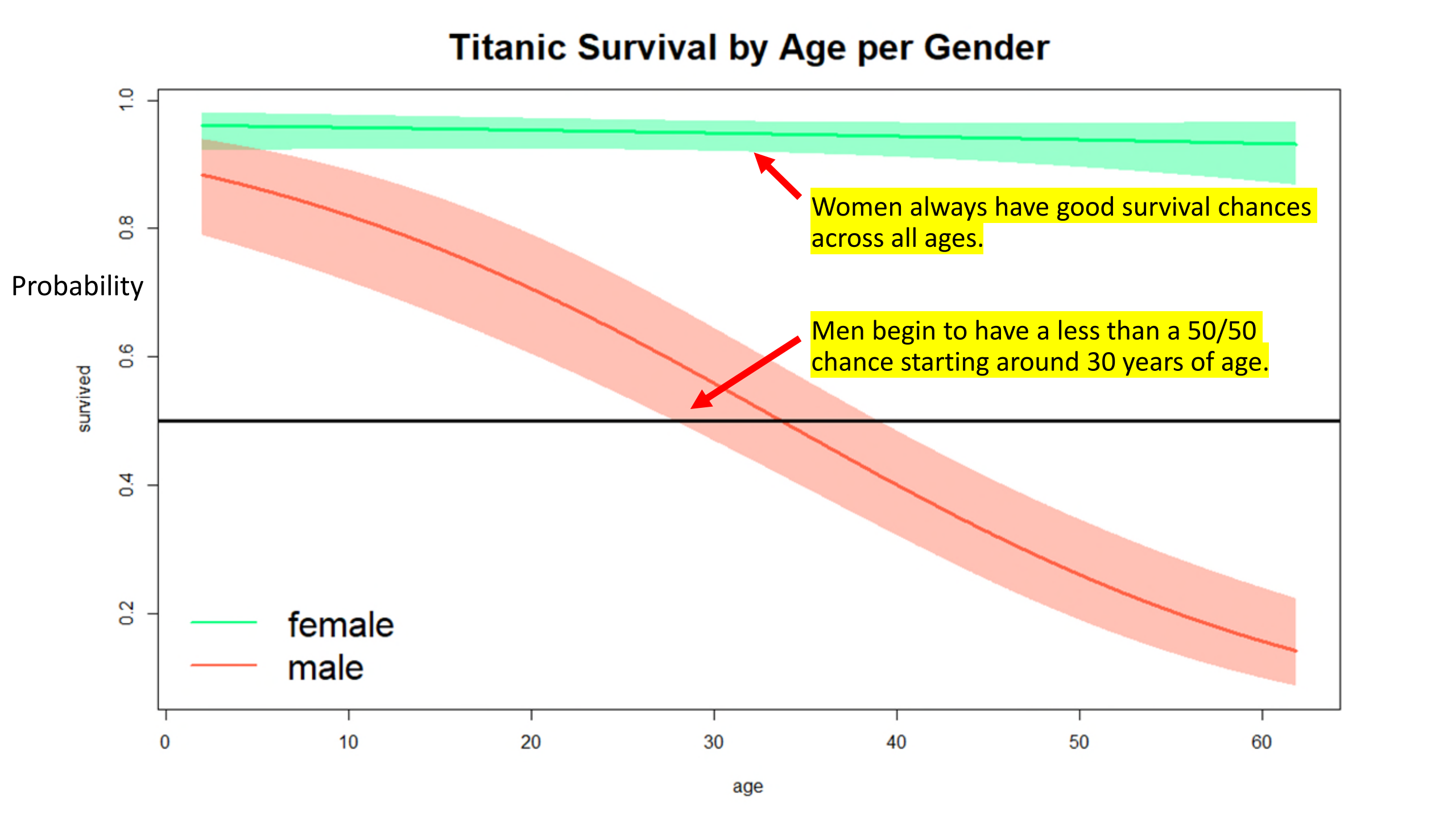
We clearly see the significant reduction in odds of surviving the Titanic. Age and sex are interactions so they depend on each other.

Probability



Significantly different probabilities of survival for each passenger class.

1st class had the best chances.


Titanic Survival by Age per Gender
Probability
survived
1.0
0.8
0.6
0.4
0.2
0
10
20
30
40
50
60
age
female
male
Women always have good survival chances across all ages.
Men begin to have a less than a 50/50 chance starting around 30 years of age.

Interpretations from the ham package.

```
Interpretations: Regression model
---------------------------------
These estimates tell you about the relationship between the
independent variables and the dependent variable. These estimates
tell the amount of change in outcome scores that would be
predicted by a 1 unit increase in the predictor.

The following predictor variable(s) have coefficient(s)
significantly different from 0 using an alpha of 0.05:
sexmale, pclass2nd, pclass3rd, sibsp, age:sexmale

For every 1 unit increase in these predictor variables,
survived is predicted to decrease by the value of the
coefficient, holding all other variables constant. The following
predictor variable(s) have negative coefficient(s) that
decrease the predicted value and odds of the outcome:
sexmale (65.3% decreased odds)
pclass2nd (78.2% decreased odds)
pclass3rd (91.4% decreased odds)
sibsp (27.5% decreased odds)
age:sexmale (5.28% decreased odds)

C-statistic/AUC. The probability of concordance is identical
to the AUC. It computes all possible pairs of subjects
(response=1 vs. response=0). The index is the proportion of such
pairs with the responder having a higher predicted probability of
response than the nonresponder. The C-statistic/AUC is 0.852.
```

Because all 4 predictors have negative effects, it is easier to interpret results as decreased chances of survival.

R and Python code used for the logistic regression model (“ham” functions in Week 5.R)

Base R

```
# object for the model
lmodel1 <- glm(survived ~ age + sex + pclass, data=titanic3)

# Summary of the model results
summary(lmodel1)
```

Python with libraries

```
lmodel1 = smf.glm('survived ~ age + sex + pclass', data=titanic3, family=sm.families.Binomial()).fit()

# Summary of the model results
print(lmodel1.summary())
```

Summary

- Millvina Dean was correct in her hypotheses.
- Logistic Regression allows us to control for multiple variables. Our **“model”** is the collection of various independent variables.
- There will never be another Titanic but her sister-ship Britannic sank 4 years later. The 30 who died would have been predicted most likely to survive, their lifeboats were pulled under (i.e., they are different samples).

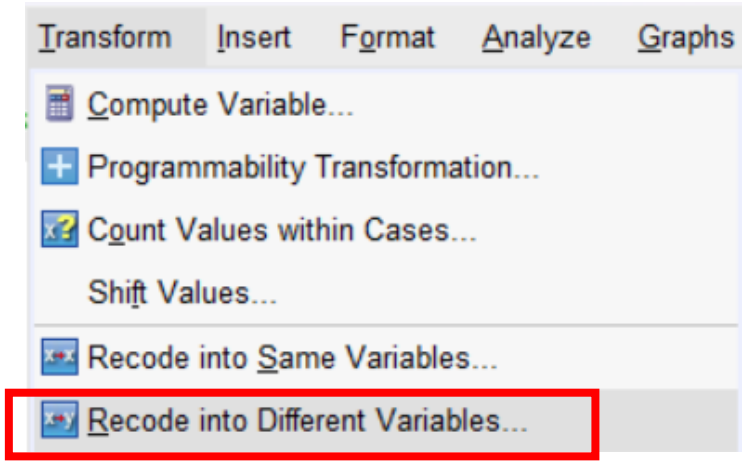
Glossary

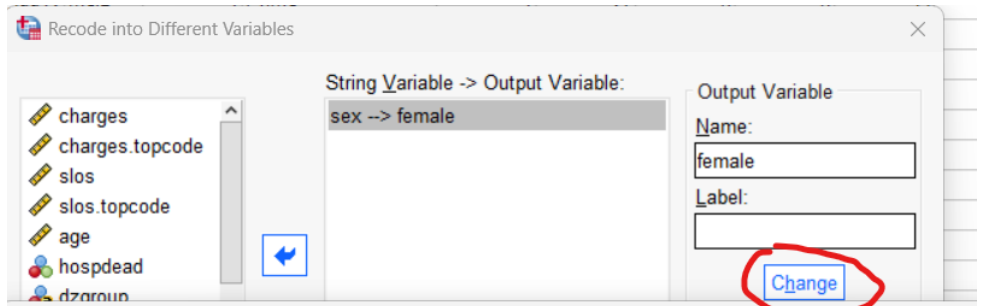
- Regression: The process of defining an outcome by one or more predictive variables. For continuous outcomes, we often use the "ordinary least squares" method to calculate regression coefficients. Also referred to as simple linear or multiple linear regression.
- Model: The set of variables used to explain the outcome of interest. For example, life expectancy might be modeled as:
  Years lived = Gender + health status.
- Coefficients: These are the regression weights used to calculate model predictions. For example, in a model that predicts weight, a height coefficient of 2 indicates that for every added inch in height, it adds 2 pounds to predicted weight.
- "Y": Outcome variable, response variable, or dependent variable. If Y is continuous, do a linear regression. If Y only has 2 levels, do a logistic regression.
- "X": Predictor variable(s), explanatory variable(s), or independent variable(s).
- R-Squared (R^2, R2): The proportion of variance of the outcome accounted for by the predictors, model reliability. i.e., how well the trend line aligns to the actual data points. More alignment= higher R^2.

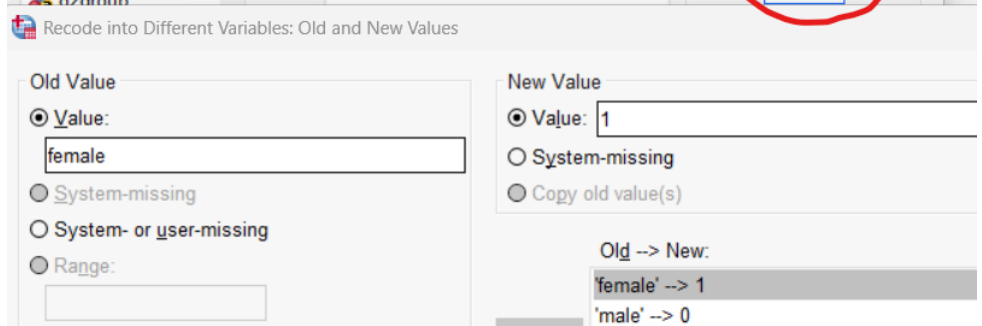
Appendix A

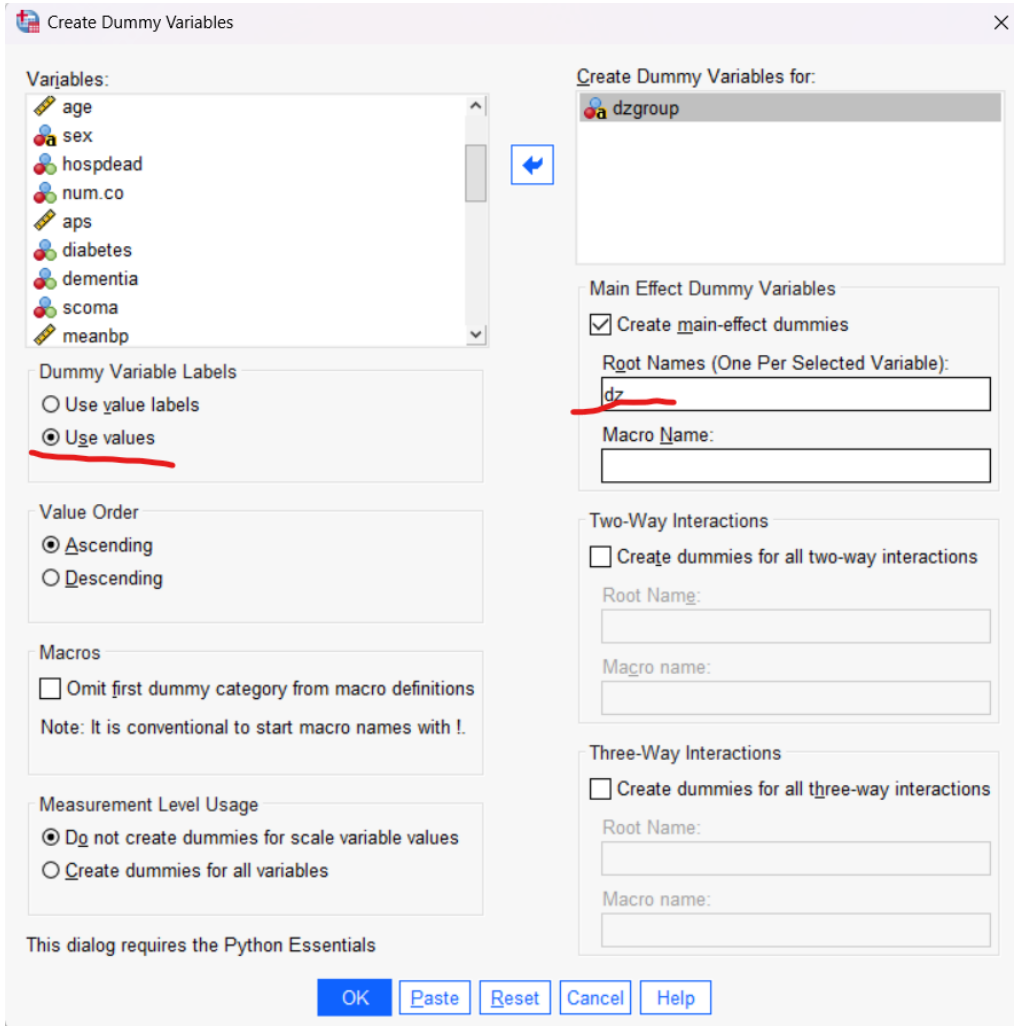
SPSS code for Linear and Logistic Regression

Linear regression: First recode or create dummy coding for sex and dzgroup





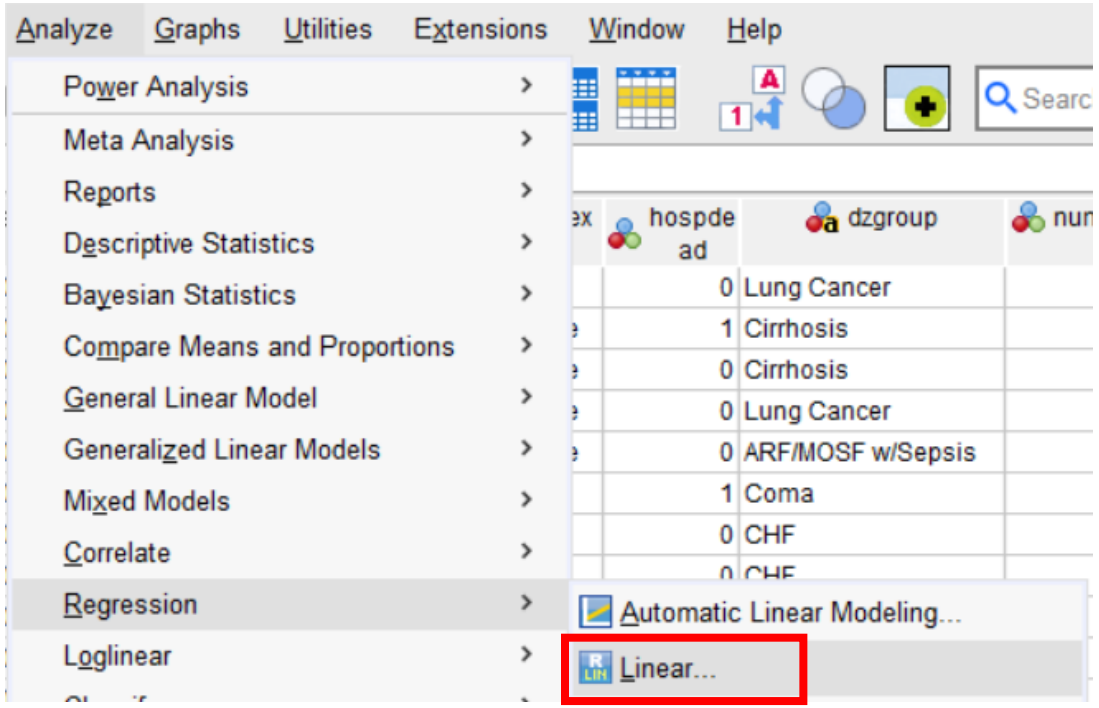


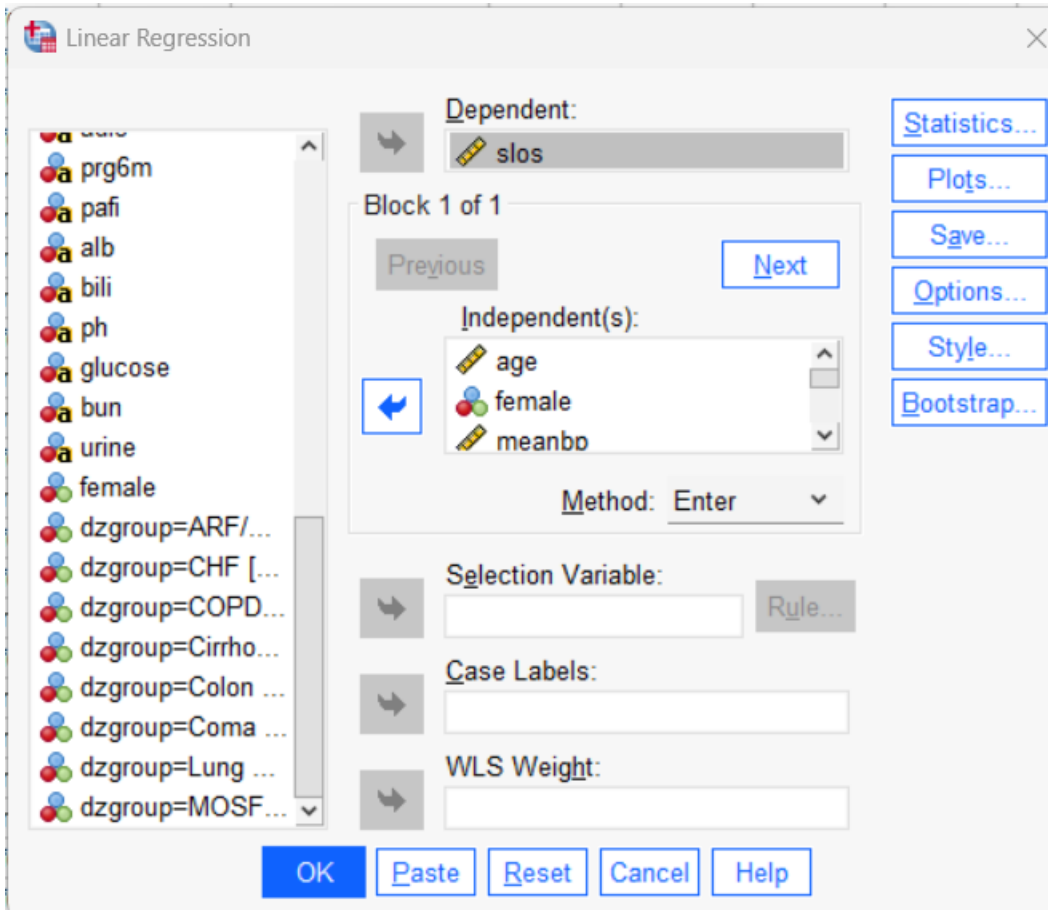


Variable Creation

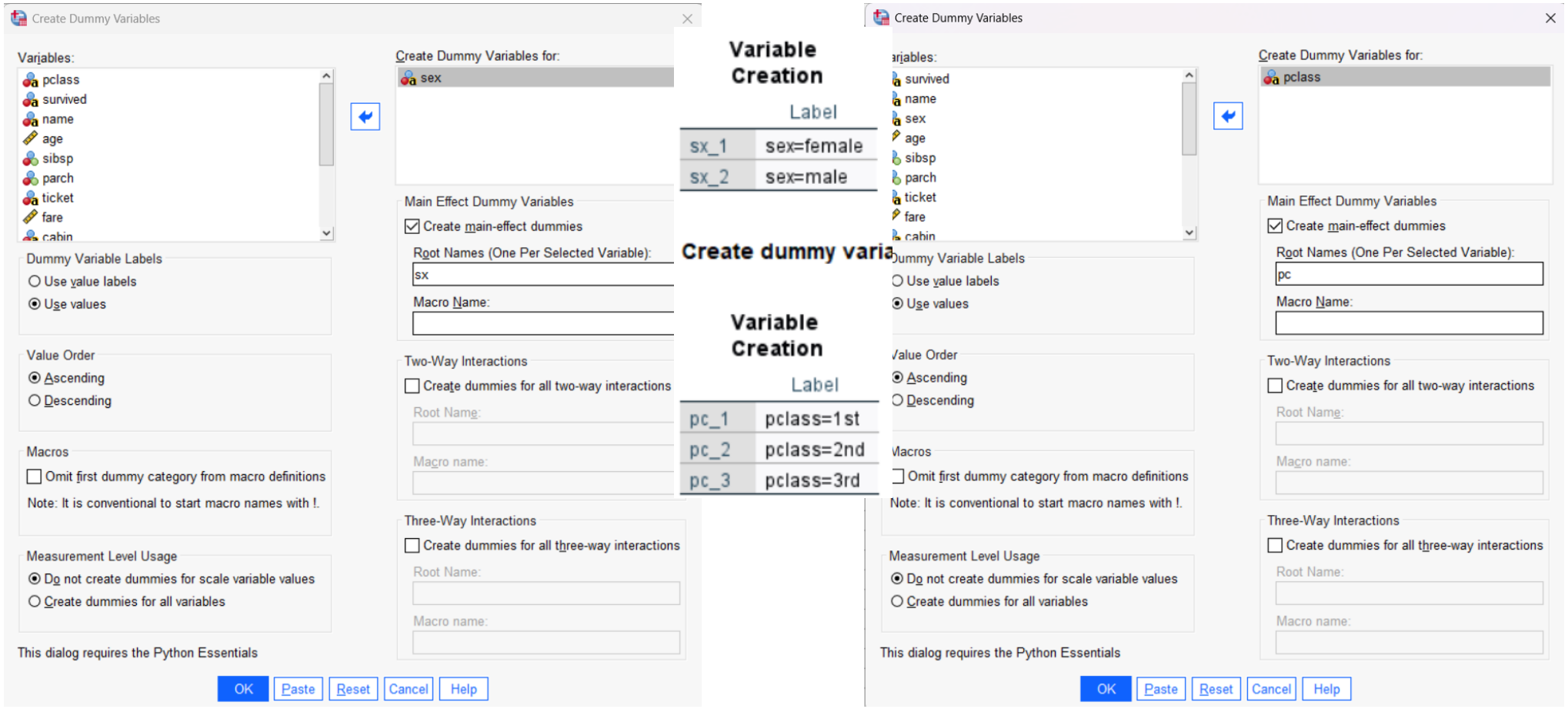
	Label
dz_1	dzgroup=ARF/ MOSF w/Sepsis
dz_2	dzgroup=CHF
dz_3	dzgroup=COP D
dz_4	dzgroup=Cirrh osis
dz_5	dzgroup=Colon Cancer
dz_6	dzgroup=Com a
dz_7	dzgroup=Lung Cancer
dz_8	dzgroup=MOSF w/Malig

Linear regression





Logistic regression: First dummy code sex and pclass



Logistic regression

